

Smart logistics is a decision loop

Digital Transformation of Modern Supply Chains · Professor Wai Kin (Victor) Chan

30

submissions

11

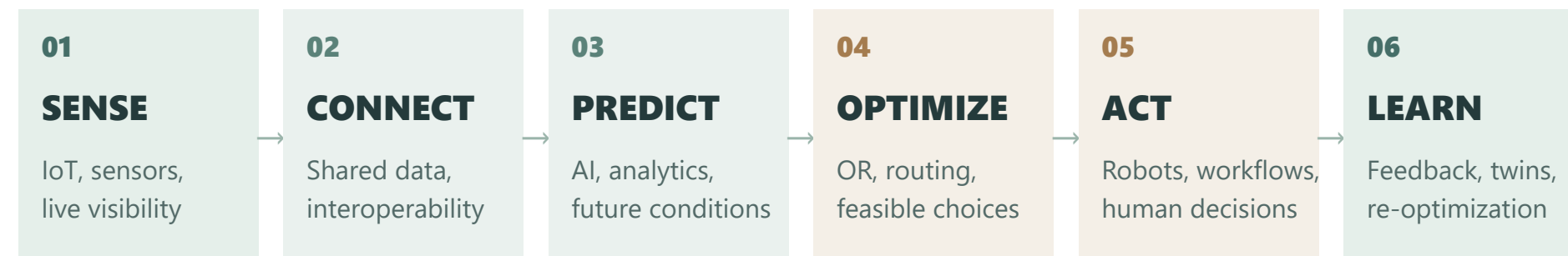
example geographies

6

linked stages

A CLASS-BUILT OPERATING MODEL

Observe → decide → execute → learn



17 paired AI + optimization

15 paired IoT + digital twin

THE GUARDRAILS AROUND THE LOOP

PEOPLE

Skills, judgement, supervision

GOVERNANCE

Privacy, safety, responsibility

RESILIENCE + SUSTAINABILITY

Limits, disruption, resource use

“Start with data, not tools.”

“AI predicts; Operations Research decides.”

WHAT Q3 MENTIONED

Technology and systems

AI

24/30

Optimization

22/30

IoT

20/30

Automation

19/30

Digital Twin

17/30

Big Data

15/30

A shared vocabulary with several connected layers.

THE NEXT 5–10 YEARS

The class expects a wider system shift

01 Decision intelligence

Prediction joined to optimization and adaptive action

02 Physical execution

Sorting, robotics and selective autonomous delivery

03 Shared foundations

Interoperable data, infrastructure and digital twins

04 Human direction

Reskilling, safety, accountability and realistic limits

EXAMPLES STUDENTS BROUGHT INTO THE ROOM

Nine snapshots across different systems and regions

VIETNAM

Viettel Post sorting

Warehouse automation

CHINA

JD Logistics delivery

Autonomous delivery

INDONESIA

Cikarang 5G warehouse

Connectivity + AGVs

MOROCCO

Tanger Med assistants

Port + AI services

CHILE

San Antonio port

Terminal + rail links

BRAZIL

APM Suape terminal

Port electrification

RUSSIA

Yandex delivery robots

Last-mile robotics

AUSTRIA

Post Vienna-Inzersdorf

Sorting operations

BANGLADESH

National Logistics Policy

Policy / infrastructure